

Customer Support Operations Platform

Milestone 1

1. Business Context

The company wants the first working foundation for the customer and support experiences.

Both applications must represent the same users and business data while providing different experiences according to the user's role.

2. Business Goals

- Allow customers and support employees to sign in.
- Represent customer, agent, and manager responsibilities.
- Connect both applications to the same business data.
- Protect application areas according to user responsibility.
- Establish a consistent application structure and navigation.

3. Users

Customers use the React application.

Support agents use the Angular application.

Support managers use the Angular application with additional responsibilities.

4. Main Business Process

1. A user signs in.
2. The system identifies the user and their role.
3. The user enters the appropriate application experience.
4. Protected areas remain unavailable to unauthorized users.
5. Both applications use the same shared business concepts and data.

5. Business Requirements

- Users must be able to sign in and sign out.
- The system must distinguish customers, support agents, and support managers.
- Both applications must use the same role and user model.
- Protected application areas must require authentication.
- Customer and support experiences must remain separated appropriately.
- A support agent must not automatically receive every manager capability.
- Protected information must remain protected even when a user changes a URL or attempts direct access.
- Both applications should provide a usable application shell with navigation and user identity.
- Authentication loading/initialization must be handled clearly.
- The product should use consistent language for requests, urgency, status, customer, and assigned employee.

6. Business Constraints

- Security must not depend only on hiding navigation or buttons.
- Both applications must operate on the same business data.
- Configuration must not expose sensitive values unnecessarily.
- The intern should document assumptions about the user and role model.

7. Delivery Expectations

Before major implementation, provide:

- clarification questions,
- selected backend/data service,
- proposed user and role model,
- proposed support-request model,
- a short explanation of how protected data will be protected.

The milestone delivery should include:

- both runnable applications,
- shared backend/data setup,

- test users,
- local run instructions,
- meaningful Git history.

8. Review Preparation

Be ready to explain:

- where authentication state is managed,
- where authorization is enforced,
- how both applications use the same user model,
- what data is safe to expose to browser code.

9. Optional Business Enhancements

Only after the milestone is complete:

- Profile editing.
- Password reset.
- Theme preference.
- Basic application branding.